JavaScript Day 4 — DOM & Events

Quick revision notes — keep this open while practicing.

1. DOM

DOM = Document Object Model. The browser turns HTML into objects that JavaScript can find, change, create and remove.

```
HTML → Browser → DOM → JavaScript controls the webpage
```

2. Select elements

Use these to find HTML elements.

```
let heading = document.getElementById("title");
let message = document.querySelector(".message");
```

3. querySelector selectors

It uses CSS-style selectors.

```
#title → ID
.message → class
p → tag
document.querySelector("#title");
document.querySelector(".message");
```

4. Change text

textContent changes text inside an element.

```
let heading = document.getElementById("title");
heading.textContent = "Welcome!";
```

5. Change styles

Use element.style to change CSS.

```
heading.style.color = "red";
heading.style.fontSize = "40px";
heading.style.backgroundColor = "yellow";
```

6. Change attributes

setAttribute changes an attribute; getAttribute reads it.

```
let photo = document.getElementById("photo");
photo.setAttribute("src", "new.jpg");
let source = photo.getAttribute("src");
```

7. Create an element

createElement makes a new HTML element.

```
let paragraph = document.createElement("p");
paragraph.textContent = "Hello!";
```

8. Add an element

appendChild puts an element into the webpage.

```
document.body.appendChild(paragraph);
```

9. Remove an element

remove takes an element off the webpage.

```
paragraph.remove();
```

10. Events

An event is something that happens on a webpage.

```
click → user clicks
input → user types/changes input
change → value/selection changes
submit → form is submitted
```

11. addEventListener — MOST IMPORTANT

It tells JavaScript to wait for an event, then run code.

```
element.addEventListener("click", function() {
// code to run after click
});
```

12. Click event

```
let button = document.getElementById("btn");
button.addEventListener("click", function() {
console.log("Button clicked!");
});
```

13. Input event

value means the current text/value inside an input.

```
let nameInput = document.getElementById("name");
nameInput.addEventListener("input", function() {
console.log(nameInput.value);
});
```

14. Change event

```
let department = document.getElementById("department");
department.addEventListener("change", function() {
console.log(department.value);
});
```

15. Submit event

preventDefault stops the browser's normal form action so JavaScript can handle it.

```
let form = document.getElementById("myForm");
form.addEventListener("submit", function(event) {
event.preventDefault();
console.log("Form submitted!");
});
```

16. To-Do List flow

1. Select input, button and list
2. Listen for button click
3. Read taskInput.value
4. Create a
5. Put task text inside
6. appendChild() to the
7. Remove task when needed

17. Final memory sheet

```
FIND → getElementById(), querySelector()
CHANGE → textContent, style, setAttribute()
CREATE → createElement()
ADD → appendChild()
REMOVE → remove()
LISTEN → addEventListener()
EVENTS → click / input / change / submit
VALUE → input.value
```